

Amy Neustein, Ph.D.
CEO and Founder
Linguistic Technology Systems
Fort Lee, NJ
917-817-2184
amy.neustein@verizon.net

Class Libraries and Full-Text Search/Encoding Tools for Publishing Documents, Datasets, and Multimedia Resources



Executive Summary ▶▶▶

PacTk is a customizable **PDF** viewer and a class-library for mid-stream publishing workflows

Use cases include:

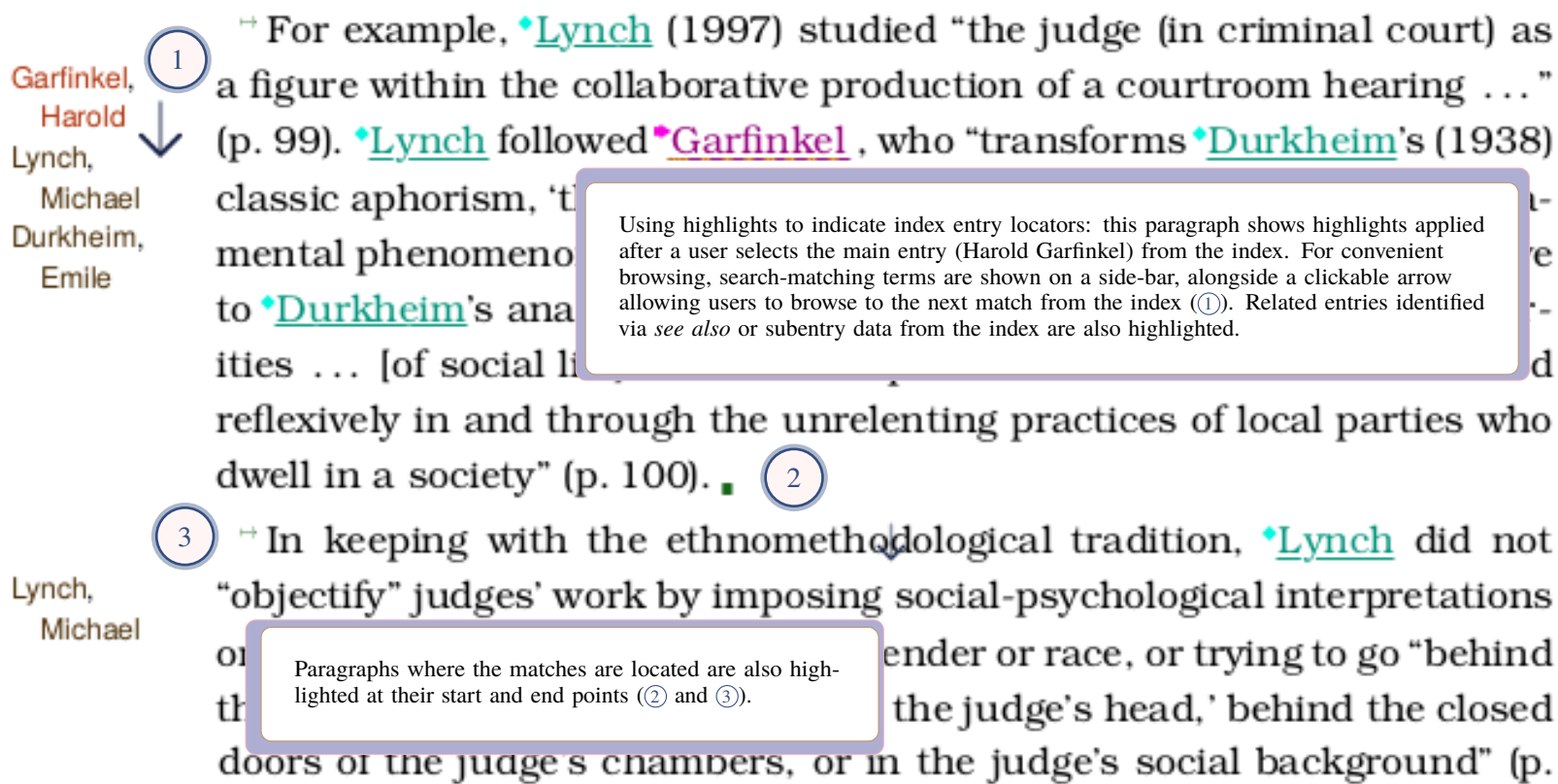
- **Navigation Handles for Finding Index Terms:** After a user selects an index entry, buttons will appear in the margins. These buttons will allow users to scroll between the pages where the index term appears, without needing to go back to the index itself each time to find the next page with a paragraph that matches the index entry (Figure 1).
- **Meta-Index for Side-by-Side Reading of Multiple Books:** This allows indexes from multiple books to be pooled into a single searchable database. Then, users who are reading one book from the publisher can select a word or phrase — the **PDF** viewer will then find matching pages from other books by the publisher and display them in a new window on the same screen so they could be read side-by-side (Figure 2).
- **Compiling Indexes:** The **PacTk PDF** viewer has special windows for indexing tasks. For example, after compiling a list of index entries, authors or production editors could instruct the **PDF** viewer to work through these index entries to find the matching words or phrases within the manuscript. The **PDF** viewer would then insert a button next to each matching paragraph that could be clicked to automatically add the current page number or Para ID to the index. Another example is that when building an index for a second edition, this matching process would be performed within both books, allowing authors and production editors to compare the results side-by-side (Figure 3).

Typically, if a paragraph was referenced in the index for the first edition, and a similar or identical paragraph is preserved in the second edition, the author will likely want the corresponding Para ID to be included in the second edition's index as well. For this reason, examining the two editions side-by-side can speed up the index process, allowing the author or production editor to conclude quickly that a second-edition paragraph matches — and hit the “yes” button to confirm, so that the locator (either as a Para ID or as a page number) will be added to the index.

- **Multimedia Plugins:** Another feature of **PacTk's PDF** viewer is the option to embed windows showing different kinds of multimedia content that can enhance or accompany scientific publications. These supplemental materials might include videos, digital maps, computer simulations, **3D**/panoramic photography, and other kinds of **3D** graphics. If a **PDF** document contains conventional **2D** graphics — for example, a map centered on some specific latitude/longitude coordinates, or a video frame, or one shot from a simulation — then the **PDF** window could be paired with a **GUI** display showing a full map, video, or **3D** graphics from which the still image is taken.

For illustration, in Figure 4, an article on organic chemistry is paired with a window rendering a molecular visualization of one compound discussed in the article.

Navigation Handles for Finding Index Terms



Garfinkel, Harold
Lynch, Michael
Durkheim, Emile

1

For example, **Lynch** (1997) studied “the judge (in criminal court) as a figure within the collaborative production of a courtroom hearing ...” (p. 99). **Lynch** followed **Garfinkel**, who “transforms **Durkheim’s** (1938) classic aphorism, ‘the mental phenomena of social life are not only determined by social conditions ... [of social life] but also reflexively in and through the unrelenting practices of local parties who dwell in a society’” (p. 100).

2

3

Lynch, Michael

In keeping with the ethnomethodological tradition, **Lynch** did not “objectify” judges’ work by imposing social-psychological interpretations on the judge’s work, or trying to go “behind the judge’s head,” behind the closed doors of the judge’s chambers, or in the judge’s social background” (p. 100).

Using highlights to indicate index entry locators: this paragraph shows highlights applied after a user selects the main entry (Harold Garfinkel) from the index. For convenient browsing, search-matching terms are shown on a side-bar, alongside a clickable arrow allowing users to browse to the next match from the index (①). Related entries identified via *see also* or subentry data from the index are also highlighted.

Paragraphs where the matches are located are also highlighted at their start and end points (② and ③).

Figure 1: Index Search in a PDF Front-End Context

Meta-Index for Side-by-Side Reading of Multiple Books

HomeSearchIndexTitlesAPIEmbedQueriesUsersPacTk



Oxford University Press

Series on Interpersonal Violence

META-INDEX — WEB ENTRY POINT

gender-based violence

LAST UPDATE: MAY 19, 2025
TITLES: 25

Page: 292 206 Search text: Pages Clear Scale PDF Document: 100%

FROM MADNESS TO MUTINY

C7P20

The difficulty of evaluating children's disclosures accurately is well recognized in the relevant literature. In 1987, in the *Journal of Interpersonal Violence*, psychiatrist David Corwin, along with other researchers, pointed out "the well-known tendency of abused children to utilize denial and dissociation in defending against their memories of abuse [which] could create a range of unpredictable and seemingly paradoxical reactions."¹²

C7P21

Roland Sur...
fornia (Los A...
modation Sys...
to sexual abus...
self-blame. D...
dren may "re...
affection towa...

C7P22

Researcher...
cases of child...
subsequent guilty plea or conviction (or by highly consistent medical evidence), found that 72 percent of the child victims actually denied the abuse when they were first questioned about it. The Arizona Coalition against Domestic Violence, in its 2003 study of protective mothers, cites the Palmer and Brown study to show that "victims don't disclose [abuse] because of fear of the consequences, self-blame, lack of awareness and difficulty in talking about the abuse."¹⁴

C7P23

Yet honest family court judges are expected to sift these paradoxical, inconsistent facts and statements and make decisions that will radically alter a family's life. It must be admitted that the judge's task in such cases is not enviable.¹⁵

C7P24

Compared with such daunting responsibility, it is fatally easy to blame the accuser. This not only disposes of the case—obviously there is no

View in Index
Meta-Index Search
Copy
Annotate/Comment

A meta-index search routed from a local PDF file through a web portal, resulting in the retrieval of another book title in the same series, or potentially anywhere within the publisher's repository. The new book is then retrieved as a PDF file and opened to the relevant page-match and can be seen side-by-side, thus keeping the reader's attention instead of opening a new browsing session.

Evan Stark, *Coercive Control: How Men Entrap Women in Personal Life* (Oxford University Press, 2023), p. 69.

Figure 2.1. Intimate partner homicide by victim gender.
Sources: By James Alan Fox & Emma E. Fridel | September 22, 20 Gender and Policy Report.

2,000
1,500
1,000
500
0

1976 1979 1982 1985 1988 1991 1994 1997 2000 2003 2006 2009 2012 2015 2018

Year

Female Male

likely as white men to be killed by their female partners. By 2004, these ratios had dropped... These changes refl... intimates dr... 1976 and 2004, the nu... males by 55%. But for the largest group of victims, white females, the number killed by an intimate dropped by only 5% in this period, and the risk to never-married white women actually increased after 1976 and has only declined very slightly since 2003. The risk to spouses of both races was higher than to boyfriend/girlfriends in 1976. But this pattern had been reversed by 2004, when the risk to boyfriend/girlfriends was considerably higher than for spouses.¹⁴ Since the domestic violence revolution began, black men have accounted for 88%

Figure 2: Meta-index search through a web portal

Compiling Indexes

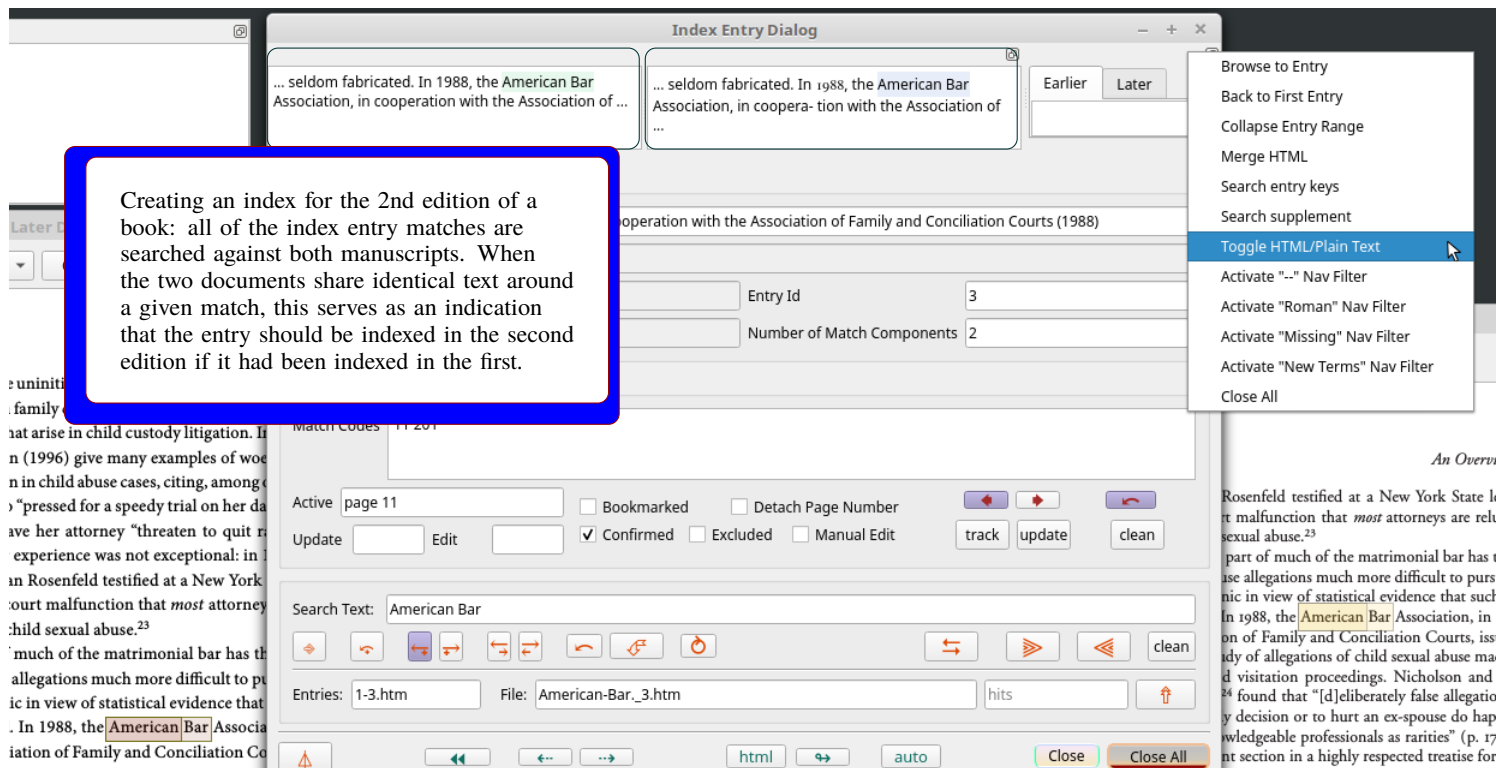
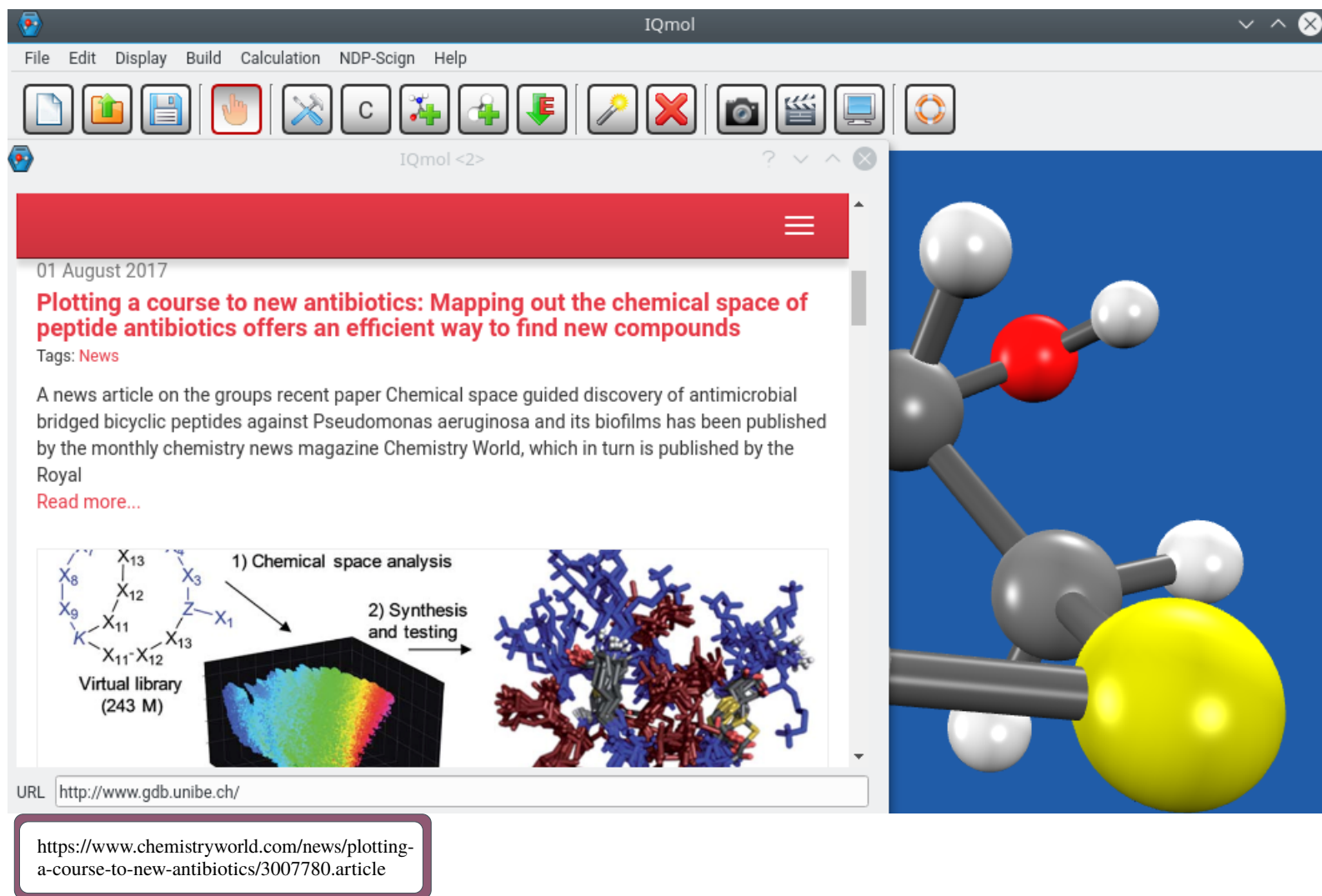


Figure 3: Compiling an index for the second edition of a book



The screenshot displays the IQmol software window. The main menu includes File, Edit, Display, Build, Calculation, NDP-Scign, and Help. The toolbar contains various icons for file operations, editing, and visualization. The central pane shows a PDF document with a news article dated 01 August 2017. The article title is "Plotting a course to new antibiotics: Mapping out the chemical space of peptide antibiotics offers an efficient way to find new compounds". The tags are News. The article text describes a recent paper on the chemical space guided discovery of antimicrobial bridged bicyclic peptides against *Pseudomonas aeruginosa* and its biofilms, published in Chemistry World. A "Read more..." link is provided. Below the text is a diagram illustrating the process: 1) Chemical space analysis of a virtual library (243 M) leading to 2) Synthesis and testing. The diagram shows a chemical structure, a 3D molecular model, and a 3D molecular visualization window on the right side of the interface.

01 August 2017

Plotting a course to new antibiotics: Mapping out the chemical space of peptide antibiotics offers an efficient way to find new compounds

Tags: News

A news article on the groups recent paper Chemical space guided discovery of antimicrobial bridged bicyclic peptides against *Pseudomonas aeruginosa* and its biofilms has been published by the monthly chemistry news magazine Chemistry World, which in turn is published by the Royal

[Read more...](#)

1) Chemical space analysis

2) Synthesis and testing

Virtual library (243 M)

URL <http://www.gdb.unibe.ch/>

<https://www.chemistryworld.com/news/plotting-a-course-to-new-antibiotics/3007780.article>

Figure 4: A PDF Plugin Showing an Interactive 3D Molecular Visualization Window